Continuity on a set

Let $f : \mathbb{R} \to \mathbb{R}$ be a real-valued function, and let $I \subseteq \text{dom}(f)$. If f is continuous at every point $x \in I$, then we say that f **is continuous on the set** I.

Proposition 5.4: All the elementary functions that we've seen in Section 4.6 are continuous on their entire domains.

Proof. We have just shown this for the cosine. The sine function is proved in similar fashion. For the other elementary functions (powers, polynomials, rational functions, the other trigonometric functions and their inverses, exponential and logarithms) we postpone the proof until a later time. □

5.4.3 Points of discontinuity

Discontinuity

Let $f : \mathbb{R} \to \mathbb{R}$ be a real-valued function. If f is not continuous at x_0 then we say that it is **discontinuous at** x_0 and x_0 is called a **point of discontinuity**.

As we shall now see, we characterize several types of discontinuities.

I. Simple discontinuities

A. Removable discontinuity

The simplest kind of discontinuity occurs when the limit $\ell = \lim_{x \to x_0} f(x)$ exists, but either $f(x_0) \neq \ell$ or $f(x_0)$ is not defined (i.e. x_0 isn't in the domain of f). In this case we can only say that f has a limit as $x \to x_0$, but that limit does not equal $f(x_0)$ (either because $f(x_0)$ is a different value, or because it is not defined).

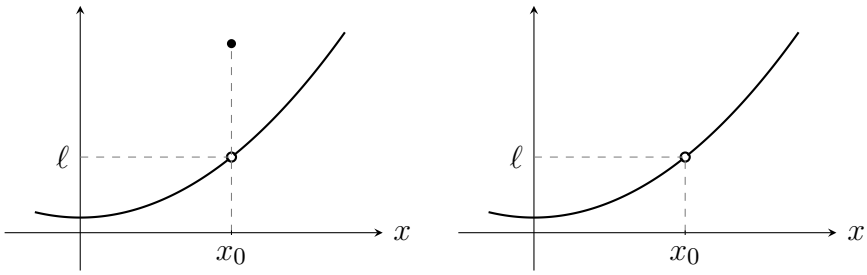


Figure 5.3: Left: $f(x_0) \neq \ell$, and right: f not defined at x_0.

Removable discontinuity

Let $f : \mathbb{R} \to \mathbb{R}$ be a real-valued function defined in a deleted neighborhood $I_r^*(x_0)$ of some point $x_0 \in \mathbb{R}$, and suppose that

$$\lim_{x \to x_0} f(x) = \ell \in \mathbb{R}$$

exists. If $f(x_0) \neq \ell$, or if f is not defined at x_0, then we say that f **has a removable discontinuity at** x_0.

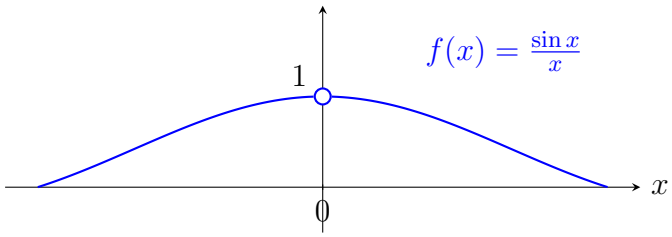
Example 5.11: We will show later that

$$\lim_{x \to 0} \frac{\sin x}{x} = 1$$

even though this function is not defined at $x = 0$. That is, $x = 0$ is a removable discontinuity of the function $\frac{\sin x}{x}$. This means that the function

$$f(x) = \begin{cases} \frac{\sin x}{x} & x \neq 0 \\ 1 & x = 0 \end{cases}$$

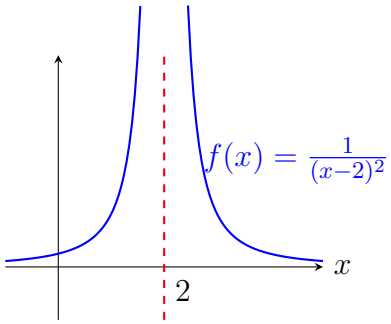
is a continuous function on $\mathbb{R}$ (we're using here the fact that $\frac{\sin x}{x}$ is continuous at all $x \neq 0$, though we have not proved that). For this reason, when mathematicians normally think about the function $\frac{\sin x}{x}$ they actually think about the function $f(x)$ which is defined at $x = 0$ as well.



B. Vertical asymptotes and jumps

Infinite limits (vertical asymptotes)

The limit of a function $f(x)$ as $x \to x_0$ need not be finite. For example, the function $f(x) = \frac{1}{(x-2)^2}$ is not defined at $x = 2$, and as $x \to 2$ the values can become arbitrarily large:



In this case we say that the function has a *vertical asymptote* at $x = 2$.

Positive infinite limit at x_0 (vertical asymptote)

Let $f : \mathbb{R} \to \mathbb{R}$ be a real-valued function defined in a neighborhood of some point $x_0 \in \mathbb{R}$, but possibly not at x_0 itself. We say that f **tends to** $+\infty$ **as** $x \to x_0$ if for every $A > 0$ there exists $\delta = \delta(A) > 0$ such that for all $0 < |x - x_0| < \delta$, we have $f(x) > A$, and we write

$$\lim_{x \to x_0} f(x) = +\infty.$$

The condition for convergence can be written symbolically as:

$$\forall A > 0,\ \exists \delta > 0,\ \text{s.t. } \forall 0 < |x - x_0| < \delta,\ f(x) > A.$$

In this case we say that the line $x = x_0$ is a **vertical asymptote** of $f(x)$.

Let us prove that this is indeed the case for $f(x) = \frac{1}{(x-2)^2}$:

Example 5.12: Show that $f(x) = \frac{1}{(x-2)^2}$ tends to $+\infty$ as $x \to 2$.

We want to show that for every $A > 0$, there exists $\delta > 0$ such that:

$$0 < |x - 2| < \delta \Rightarrow \frac{1}{(x-2)^2} > A.$$

Fix $A > 0$. We need to find $\delta > 0$ such that:

$$\frac{1}{(x-2)^2} > A$$

This inequality is equivalent to:

$$(x-2)^2 < \frac{1}{A}$$

Taking square roots (and noting both sides are positive):

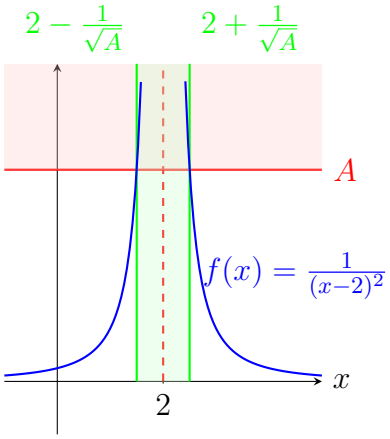
$$|x - 2| < \frac{1}{\sqrt{A}}$$

Therefore, if we choose $\delta = \frac{1}{\sqrt{A}}$, then for $0 < |x - 2| < \delta$:

$$\frac{1}{(x-2)^2} > \frac{1}{\delta^2} = A.$$

Since $A > 0$ was arbitrary, we conclude that

$$\lim_{x \to 2} \frac{1}{(x-2)^2} = +\infty.$$



Negative infinite limit at x_0 (vertical asymptote)

Let $f:\mathbb{R}\to\mathbb{R}$ be a real-valued function defined in a neighborhood of some point $x_0\in\mathbb{R}$, but possibly not at x_0 itself. We say that f **tends to** $-\infty$ **as** $x\to x_0$ if for every $A>0$ there exists $\delta=\delta(A)>0$ such that for all $0<|x-x_0|<\delta$, we have $f(x)<-A$, and we write

$$\lim_{x\to x_0} f(x)=-\infty.$$

The condition for convergence can be written symbolically as:

$$\forall A>0,\ \exists\delta>0,\ \text{s.t. } \forall 0<|x-x_0|<\delta,\ f(x)<-A.$$

In this case we also say that the line $x=x_0$ is a **vertical asymptote** of $f(x)$.

Note that as defined here, the infinite limit must be the same whether x tends to x_0 from the right or from the left. Hence, the function $f(x)=\frac{1}{x}$ does *not* have a limit as $x\to 0$.

Jump discontinuities: different limits on the left and on the right

We have already seen some examples of functions that have points where we tend to different values if we approach from the left or from the right. Two simple examples at the functions $f(x)=\frac{1}{x}$ and the ceiling function $g(x)=\lceil x\rceil$, sketched below.

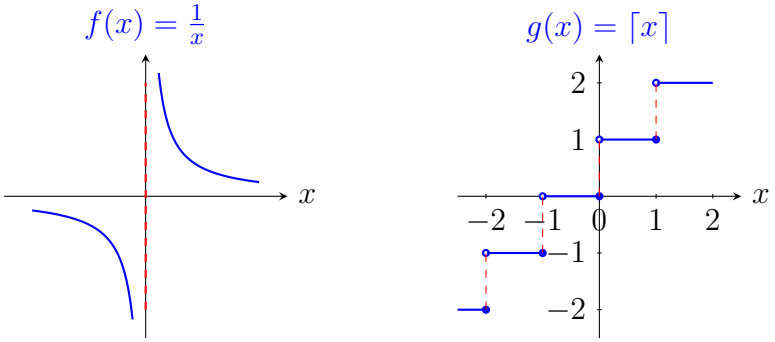


Figure 5.4: Examples of functions that have points with different left and right limits.

Hence we want to repeat the ideas that we've seen above, with the only difference being the neighborhoods around x_0: we'll want a *right* neighborhood and a *left* neighborhood (see Figure 5.5).

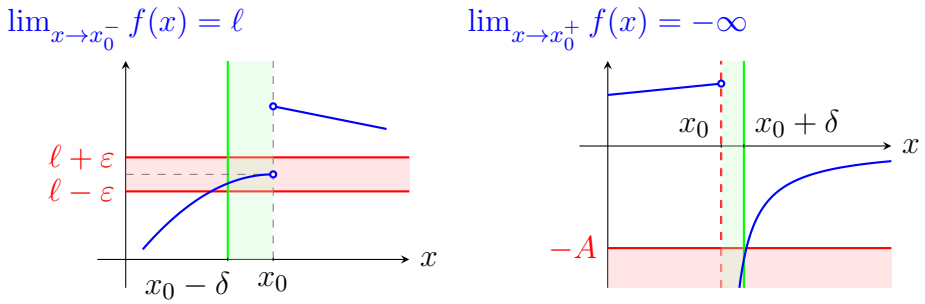


Figure 5.5: Left: a left neighborhood $(x_0 - \delta, x_0)$, on which f stays within ε of its left limit ℓ. Right: a right neighborhood $(x_0, x_0 + \delta)$, on which f stays below $-A$, so that its right limit is $-\infty$.

Finite right limit

Let $f : \mathbb{R} \to \mathbb{R}$ be a real-valued function. We say that f **has a right limit** ℓ **at** $x_0 \in \mathbb{R}$ if for every $\varepsilon > 0$ there exists $\delta = \delta(\varepsilon) > 0$ such that for all $x \in (x_0, x_0 + \delta)$, we have that $f(x) \in (\ell - \varepsilon, \ell + \varepsilon)$ and we write

$$\lim_{x\to x_0^+} f(x) = \ell.$$

The condition for a right limit can be written symbolically as:

$$\forall \varepsilon > 0,\ \exists \delta = \delta(\varepsilon) > 0,\ \text{s.t. } 0 < x - x_0 < \delta \ \Rightarrow\ |f(x) - \ell| < \varepsilon.$$

Right-continuous

A function $f : \mathbb{R} \to \mathbb{R}$ is said to be **right-continuous at** x_0 if

$$\lim_{x\to x_0^+} f(x) = f(x_0).$$